

# Convolutional Bottleneck Attention for Efficient Hybrid Vision Transformers

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Code: <https://github.com/mujeebasad/cba-mobilevit>

**Abstract**—Hybrid CNN–ViT architectures for edge deployment defer self-attention until the convolutional backbone has reduced spatial extent to  $8 \times 8$  or  $16 \times 16$  tokens. The early stages stay pure convolution, where receptive field grows only sub-linearly with depth [1], so full-image context is unreachable until late in the pipeline. The natural alternative — *deep bottleneck attention*: compress aggressively, run attention on the compressed latent, decode back, and place the block at *high* spatial extent — has stayed unoccupied because of two concerns: latent geometry (a class-conditionally clustered latent might be incoherently mixed by attention) and compute cost (a multi-stage non-linear encoder at high spatial extent is too expensive).

The **Convolutional Bottleneck Attention (CBA)** block addresses both. The latent-geometry concern is treated as open: the trained network reaches 81.52% top-1, with three candidate mechanisms catalogued [2] for future verification. The compute concern is resolved by two design choices: convolution over learned projection matrices [3], [4] (preserving the CNN inductive priors that the decoder needs for a spatially-coherent gate), and within the conv family, MobileNetV2-style depthwise-separable convolution [5] ( $8\text{--}9\times$  cheaper than standard at marginal capacity loss). Together these make a two-stage non-linear encoder around attention tractable at  $128 \times 128$  for the first time.

CBA-1 sits after stage 1 of MobileViT-S at  $128 \times 128 \times 32$ , running attention on the  $16 \times 16 \times 72$  compressed latent — full-image receptive field at a depth where standard hybrids operate on  $\sim 11$ -pixel local patches. The block integrates via a per-pixel-per-channel sigmoid gate inside an additive residual,  $Y = X + \text{DropPath}(\sigma(z) \odot X)$ , which preserves the He et al. gradient highway [6] and bounds the multiplier to  $(1, 2)$ . CBA-MobileViT-S-lite (2.79 M params, 1.114 G MACs), trained from scratch on ImageNet-100 at  $256 \times 256$  for 50 epochs, reaches **81.52%** top-1 (EMA) — **+4.62 pp** over the matched-protocol MobileViT-XS at +0.18 G MACs and **+3.48 pp** over MobileViTv2-0.75 despite a schedule asymmetry that favours the latter.

**Index Terms**—Image classification; vision transformer; mobile and edge deep learning; attention mechanism; depthwise-separable convolution; residual gating; hybrid CNN–transformer architecture; bottleneck attention.

## I. INTRODUCTION

### A. Where attention sits in a hybrid mobile vision pipeline

Edge-deployed vision models operate under hard constraints (battery, thermal envelope, memory, latency) that data-center models do not. The mobile-vision literature has converged on two responses: efficient CNN families [5], [7] built on depthwise-separable convolutions, and hybrid CNN–ViT architectures [8]–[12] that keep a convolutional backbone for low-level processing and add attention for global context. The hybrid camp is the frame for this work.

Across that hybrid lineage, self-attention is introduced *only after* the backbone has reduced spatial extent to  $8 \times 8$  or  $16 \times 16$  — the last 2–3 stages of a 5-stage MobileViT-class backbone. The reason is cost-driven: standard attention scales  $O(N^2)$  in token count  $N$ , so applying it at the  $128 \times 128 = 16,384$  token resolution would cost several hundred times more than at  $8 \times 8 = 64$  tokens. Every variant takes the same way out: defer attention until the quadratic cost is manageable.

The cost is what’s deferred; the consequence is that the *early* stages of every hybrid mobile model remain pure-convolutional. Effective receptive field grows only sub-linearly with depth [1], so full-image context is unreachable until spatial extent has been reduced  $\sim 16\times$  from input. Object-spanning and scene-spanning features that need evidence from distant positions cannot be built at the early stages.

**The unexplored alternative.** Rather than “wait for the backbone to compress, then run attention on a small token set”, *compress aggressively in one block via a learned encoder, run attention on the compressed latent, then decode back*. This places a full-image receptive-field operator at high spatial extent, early in the pipeline, where standard hybrids still operate purely convolutionally. Two structural concerns have kept this design space unoccupied:

- 1) *Latent geometry.* A classification-trained encoder is widely believed to push same-class inputs together and different-class inputs apart, and attention’s softmax-weighted token-averaging is feared to mix this clustering incoherently, producing decoder garbage. Existing precedents [3], [4], [13], [14] sidestep this by keeping the bottleneck linear, compressing only channels (not space), or pretraining the encoder with auxiliary losses.

- 2) *Compute cost.* A standard  $3 \times 3$  conv at  $128 \times 128 \times 32$  costs  $\sim 100$  M MACs alone; two such stages would dominate the model budget.

The Convolutional Bottleneck Attention (CBA) block addresses both. The latent-geometry concern is treated as genuinely open: the trained model works (81.52% top-1 on ImageNet-100), and Section IV-B catalogues three literature-traceable mechanisms that may explain why. The compute concern is resolved by two ordered design decisions (Section IV-C): **convolution** over learned projection-matrix alternatives (preserving the CNN inductive priors the decoder needs for a spatially-coherent gate); within the conv family, MobileNetV2-style **depthwise-separable convolution** ( $\sim 8-9 \times$  cheaper than standard at marginal capacity loss). The architectural payoff (Section IV-D): **CBA-1 sits after MobileViT-S's stage 1 at  $128 \times 128$ , with attention computed on the  $16 \times 16 \times 72$  compressed latent** — full-image receptive field at a depth where pure CNN gives only  $\sim 11$ -pixel local features.

## B. Contributions

- 1) **Identification of deep bottleneck-attention as a previously unexplored architectural axis** for hybrid mobile vision transformers, with the two structural concerns above made explicit and existing nearby precedents located on a design-space taxonomy (Figure 1).
- 2) **An empirical-feasibility position on the latent-geometry concern** based on end-to-end self-consistency: the trained model empirically works (81.52% top-1), and three literature-traceable mechanisms are catalogued as candidate explanations whose verification is future work (Section IV-B).
- 3) **Resolution of the compute concern in two ordered design decisions:** convolution over learned-projection-matrix alternatives (preserving CNN inductive priors [3], [4]), and MobileNetV2-style depthwise-separable convolutions within the conv family [5] (Section IV-C).
- 4) **The CBA block** as the concrete instantiation: input  $\rightarrow$  MV2-DSC encoder  $\rightarrow$  MHSA on compressed latent  $\rightarrow$  MV2-DSC decoder  $\rightarrow$  per-pixel-per-channel sigmoid gate inside an additive residual, gradient-safe per He'16 [6] with effective multiplier  $1 + \alpha \in (1, 2)$ .
- 5) **CBA-MobileViT-S architecture:** 2.79 M parameters, 1.114 G MACs.
- 6) **Empirical evaluation on ImageNet-100** for 50 epochs at  $256 \times 256$ , against MobileViT-XS (matched-protocol control on the same Blackwell GPU and recipe) and MobileViTv2-0.75 (earlier run on Kaggle T4 under a more favourable schedule, methodologically disclosed). CBA-MobileViT-S-lite: 81.52% **top-1** (EMA), +4.62 pp over MobileViT-XS at +0.18 G MACs and +3.48 pp over MobileViTv2-0.75 despite the schedule asymmetry.

## II. BOTTLENECK-ATTENTION PRECEDENTS

**Mobile vision backbones.** The mobile-CNN lineage [5], [7], [15] established depthwise-separable convolution and inverted-residual blocks as the standard primitives. The hybrid

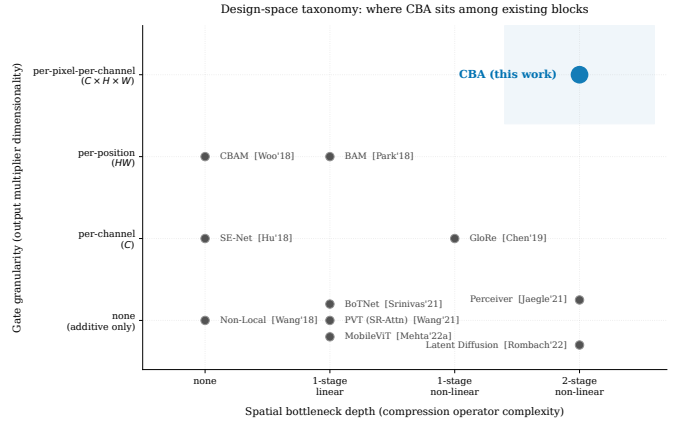


Fig. 1. Design-space taxonomy.  $x$ -axis: spatial bottleneck depth.  $y$ -axis: gate granularity. CBA (top right) is the unique block combining two-stage non-linear conv compression with per-pixel-per-channel multiplicative gating, trained end-to-end by classification loss alone.

CNN-ViT era [8]–[12] keeps a convolutional backbone for low-level processing and inserts attention only at late stages, after spatial extent has been reduced to  $8 \times 8$  or  $16 \times 16$ . This deferral is the structural choice the present work re-examines.

**Attention efficiency.** Two parallel lines reduce attention's cost. The first manipulates the token *count* — DynamicViT [16], EViT [17], ToMe [18] — by routing on signals derived from a full attention pass. The second — the lineage CBA extends — manipulates the *space* over which attention is computed by inserting a bottleneck before the attention sub-layer. The precedents most relevant to CBA:

- **PVT** [4]: Spatial-Reduction-Attention via a single strided  $1 \times 1$  conv on  $K, V$  before attention. Linear (no non-linearity), one stage.
- **Linformer** [3]:  $K, V$  projected through a learned matrix  $E \in \mathbb{R}^{N \times k}$  to a fixed-length sequence. Linear and *token-dimension* compression — no spatial inductive bias whatsoever.
- **BoTNet** [13]: replaces the  $3 \times 3$  conv in the last three ResNet bottleneck blocks with MHSA — but at *full* spatial resolution; only *channel* compression.
- **Perceiver** [14]: small set of learnable latents updated by cross-attention to the input. Standalone architecture, not an in-CNN-block residual gate.
- **BAM** [19]: bottleneck attention modules at network bottlenecks, but uses MLP-with-ReLU + dilated-conv rather than self-attention.

The taxonomy is summarised by the design-space scatter in Figure 1: spatial bottleneck depth  $\times$  gate granularity. **No existing block occupies the position** (two-stage non-linear conv bottleneck)  $\times$  (multi-head self-attention on compressed latent)  $\times$  (placed at high spatial extent)  $\times$  (trained end-to-end by classification loss alone). CBA enters this gap.

### III. PRELIMINARY INVESTIGATION: PRE-ATTENTION GATING

Before settling on the CBA design, we tried a different idea: if a cheap closed-form score could pick which tokens the transformer actually needs, we could throw the rest away pre-attention and recover compute. The result was negative. This section documents what we tried, what we measured, and what we learned — because the failure shape is what motivated the architectural pivot.

#### A. Original hypothesis

The convolutional branch of a hybrid model produces, before any attention runs, a feature map  $X \in \mathbb{R}^{B \times C \times H \times W}$  that already encodes spatial saliency. We hypothesised that a cheap, closed-form scoring function  $s: \mathbb{R}^C \rightarrow \mathbb{R}$  applied per spatial position could rank tokens by attention-relevance, letting us keep only the top  $K = \lfloor N\rho \rfloor$  at keep ratio  $\rho$  and bypass the rest through the existing residual. Token reduction in pure ViTs [16]–[18] achieves 30–50% sparsity at  $< 1$  pp loss but always *post-attention*; a free CNN-derived *pre-attention* signal would be strictly cheaper.

#### B. Experimental design

We instrumented MobileViT-S [8] with five non-learned scoring functions on the per-position channel vector  $x_{i,j} \in \mathbb{R}^C$  — normalised Shannon entropy,  $L^1$  magnitude,  $L^2$  norm, cross-channel variance, and max-channel activation — selecting the top- $K$  at keep ratios  $\rho \in \{0.3, 0.5, 0.7, 1.0\}$ . Bypassed tokens passed through MobileViT-S’s existing `conv_fusion` residual; selected tokens flowed through the unmodified transformer block.

#### C. Results

The headline result is for the entropy gate at  $\rho=0.5$  on ImageNet-100. Baseline (no gate): 91.64% top-1, 308 img/s on a T4. Zero-shot with the gate: 1.12% (chance). After 5 epochs of full-model fine-tuning: 83.64% at 392 img/s —  $-8$  pp accuracy for  $+27\%$  throughput. The other four scoring functions clustered within run-to-run noise of entropy. The trade-off is dominated by the trivial alternative of running MobileViT-XS at higher throughput with no gating logic.

#### D. Why the closed-form approach failed

A scalar reduction  $s(x_{i,j})$  of a  $C$ -dimensional channel vector at one spatial position is monotonic in some statistic of activation magnitude. The property “this token, in this layer, will benefit from full self-attention” is not such a function: it depends on which other tokens populate the sequence, which queries the learned  $W_Q, W_K, W_V$  will form, and which downstream task is being optimised. Per-position channel statistics encode none of these.

Three avenues remain to recover a workable pre-attention gate: (a) learn the gate as a sparsity-regularised scoring network [16] — forfeits the “free” signal that motivated the hypothesis; (b) gate post-attention [17], [18] — the  $O(N^2)$  cost has already been paid; (c) redesign the attention block so

gating is intrinsic to its operation rather than a pre-filter. The CBA block develops avenue (c).

### IV. DESIGN HYPOTHESIS AND EVOLUTION

#### A. Deep bottleneck attention as an unexplored axis

The lesson from Section III is that gating must be *intrinsic* to attention, not a pre-filter applied on top. The shape of that redesign is the unfilled position in Figure 1: a two-stage non-linear conv bottleneck encoder, self-attention on the compressed latent, placed early in the pipeline, and trained end-to-end by classification loss alone. Two structural concerns explain why the position has stayed unfilled — latent geometry and compute — and the next two subsections address them in order.

#### B. Concern O1: Latent geometry under task supervision

A classification-trained encoder is expected to cluster same-class inputs together and push different-class ones apart, and attention’s token-averaging is feared to mix this clustering incoherently into decoder garbage. This is the structural reason behind the otherwise curious design choices in nearby precedents: PVT keeps the bottleneck linear; BoTNet compresses only channels; Perceiver regularises the encoder with auxiliary losses.

We treat O1 as an open empirical question. What we can claim is that the architecture trained end-to-end by classification loss *works*: 81.52% top-1 on ImageNet-100, monotonic training with no rank-collapse signatures, and visibly object-localised gate maps (mean  $\bar{\alpha} \in [0.10, 0.18]$  with maxima approaching 1.0; Figure 2). The encoder, attention, decoder, and gate are jointly optimised by a single classification loss with no auxiliary objective, so the encoder’s latent is, by construction, whatever makes the downstream attention output useful for classification. This is a feasibility claim, not a mechanistic explanation of why such a fixed point exists.

Three candidate mechanisms in the literature could underlie that fixed point. First, *token identity*: attention here operates on the  $N=256$  tokens of a single forward pass — all derived from the same image — so its mixing is within-image, never across-class clusters. Class-conditional clustering is a dataset-level property, not a per-image one. Second, the *Information Bottleneck* [2]: deep VIB [2] reports class-conditional clustering at high  $\beta$  with within-class variation preserved as task-relevant signal, and if CBA’s deterministic encoder reaches a comparable point, attention operates in the preserved within-class subspace. (IB theory is stochastic, so this remains an intuition.) Third, *working precedents*: PVT [4] and BoTNet [13] both run self-attention on task-supervised compressed latents and train cleanly, and Linformer’s Theorem 1 [3] bounds attention’s effective rank to  $\Theta(\log n)$  — attending on a low-rank latent loses essentially nothing the original tokens could express. None of these is CBA, but each demonstrates the operator class is feasible.

Empirically distinguishing these mechanisms (per-image vs. across-image latent geometry; within-class vs. between-class cosine and linear probes; rank profile across depth) is future

work (Section VIII-D); the architectural contribution does not depend on it.

*C. Concern O2: Compute cost, resolved by depthwise-separable convolutions*

A standard  $3 \times 3$  conv at CBA-1’s  $128 \times 128 \times 32$  input with  $32 \rightarrow 64$  channels costs  $\sim 302$  M MACs; two such stages would cost  $\sim 600$  M — half the entire MAC budget of a MobileViT-XS-class model. This is the arithmetic that has steered every nearby precedent to a less expressive operator. We resolve it with two ordered choices: (D1) convolution over learned-projection-matrix alternatives, and (D2) within the conv family, MobileNetV2-style depthwise-separable convolution.

1) *D1. Convolution over learned-projection-matrix alternatives*: The cheaper precedents — PVT’s strided  $1 \times 1$  conv [4] and Linformer’s learned token-axis projection [3] — discard structural priors that vision needs. Linformer’s projection in particular has *no spatial inductive bias*: nothing in its learnable matrix encodes that adjacent tokens should mix more strongly than distant ones, that translation should be equivariant, or that output tokens retain spatial identity. A  $3 \times 3$  conv encodes all three structurally; a free learned matrix would have to discover them from data, unaffordable in our small-data regime ( $\sim 126$  K IN-100 images, no IN-1K pretraining, 50 epochs).

There is a second reason for conv: spatial identity in the latent. Each token at output position  $(i', j')$  corresponds to a receptive-field region of the input. This is what the decoder maps back to a spatially-coherent gate, and what makes Section IV-D’s receptive-field claim well-defined. With a Linformer-style projection, the  $k$  output tokens are abstract learned combinations with no spatial identity — there is nothing for the decoder to “decode back to”.

2) *D2. Depthwise-separable convolution over standard convolution*: A standard  $D_k \times D_k \times C_{\text{in}} \times C_{\text{out}}$  kernel factors as a depthwise spatial filter followed by a pointwise mixer:

$$K(p, q, m, n) \approx K_{\text{DW}}(p, q, m) \cdot W(m, n), \quad (1)$$

with cost ratio  $\text{MACs}_{\text{DSC}}/\text{MACs}_{\text{std}} = 1/C_{\text{out}} + 1/D_k^2 \approx 1/9$  at  $D_k=3$  — about  $8\text{--}9\times$  cheaper per layer with no classification-capacity loss at scale [7], [15].

The MV2 inverted-residual [5] refines this for the bottleneck setting: PW expand  $\rightarrow$  DW at expanded width  $\rightarrow$  linear PW project. For CBA-1’s  $128 \times 128 \times 32 \rightarrow 16 \times 16 \times 72$  encoding, the two-stage MV2-DSC encoder costs  $\sim 36$  M MACs versus  $\sim 600$  M for a standard-conv equivalent — the difference between tractable and impossible at high spatial extent. The decoder mirrors the encoder with bilinear upsampling inserted between PW-expand and DW; bilinear is artifact-free and cheaper than transposed convolution.

*D. Architectural consequence: full receptive field at high spatial extent*

With O1 and O2 addressed, the headline architectural payoff follows: **attention can be placed at high spatial extent, granting full-image receptive field at a stage where standard hybrids still operate purely convolutionally.**

CBA-1 sits after stage 1 of MobileViT-S at  $128 \times 128 \times 32$ , compresses to a  $16 \times 16 \times 72$  latent ( $s_1=4, s_2=2$ ), runs  $L=2$  MHSA layers on the resulting 256-token sequence, and decodes back to  $128 \times 128 \times 32$ . Attention’s  $O(N^2)$  cost is paid at  $N=256$  ( $\sim 65$  K pairwise interactions) — never at the  $N=16,384$  a direct  $128 \times 128$  attention would require.

In a stacked CNN with  $D_k=3$  kernels and stride-2 down-samples every other stage, the effective receptive field grows sub-linearly with depth [1]. By the end of stage 1 of MobileViT-S, each output position has only  $\sim 11$ -pixel local access — roughly 4% of input extent. CBA-1 replaces this with full-image receptive field in a single block: every position in the  $16 \times 16$  latent attends to every other, and the decoder maps the result back to  $128 \times 128$ , so every output position has received contributions from every input position via the attention pathway. CBA-2/3/4 follow the same template at progressively reduced spatial extent ( $32^2, 16^2, 8^2$ ); CBA-1 is the load-bearing instance.

Figure 2 shows the qualitative footprint: CBA-1’s gate maps are the most object-localised of the four blocks, consistent with the spatial heavy lifting happening at the early stage.

*E. Supporting choice: positional encoding is omitted*

CBA omits all explicit positional encoding. The DSC encoder’s two  $3 \times 3$  depthwise convolutions with zero-padding implicitly encode position via the PEG mechanism [20], and the literature consistently shows that explicit PE adds nothing when convolutions surround attention [8].

*F. Supporting choice: FFN is retained*

The FFN is retained at modest expansion factor  $f=2$  (vs. ViT/DeiT’s  $f=4$ ). MetaFormer [21] shows the channel MLP is the load-bearing component: IdentityFormer-M48 with no token mixing reaches 80.4% IN-1K. The FFN also raises the per-layer Lipschitz constant, slowing rank collapse in deep stacks of attention layers.

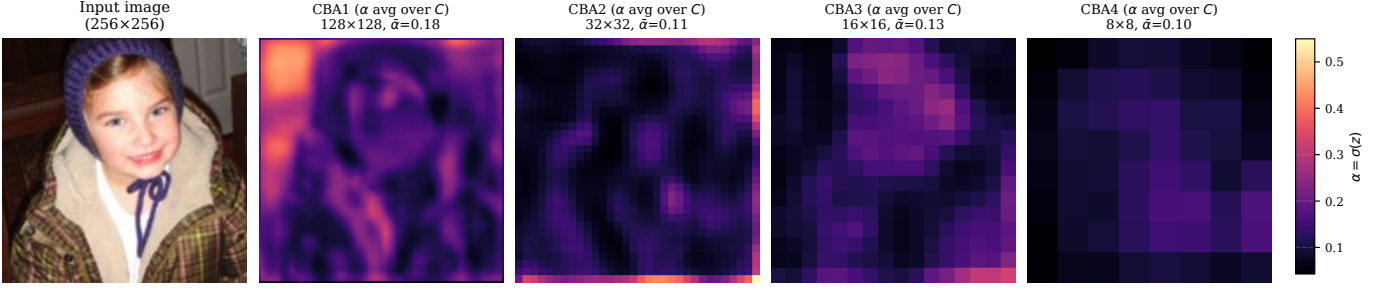
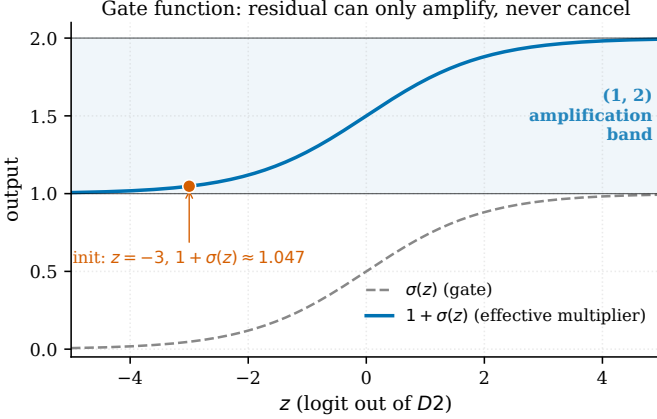
*G. Supporting choice: residual integration via Hadamard inside additive shortcut*

We integrate the post-attention output  $z$  with the input  $X$  as  $Y = X + \alpha(X) \odot X = (1 + \alpha(X)) \odot X$  with  $\alpha = \sigma(z) \in (0, 1)^{B \times C \times H \times W}$ . Three properties justify the choice.

**Proposition 1** (Hadamard-vs-additive rank growth). *For  $A, B \in \mathbb{R}^{m \times n}$ ,  $\text{rank}(A+B) \leq \text{rank}(A) + \text{rank}(B)$  (linear), whereas  $\text{rank}(A \odot B) \leq \text{rank}(A) \cdot \text{rank}(B)$  (Schur 1911 [22], [23]; quadratic).*

CBA’s residual  $\alpha \odot X$  thus has rank up to  $\text{rank}(\alpha) \cdot \text{rank}(X)$  — quadratic, strictly dominating the rank-1 (per-channel) gate of SE-Net.

**Proposition 2** (Gradient highway preservation). *For  $Y = X + \alpha(X) \odot X$  with  $\alpha \in (0, 1)^d$ ,  $\partial Y / \partial X = \text{diag}(1 + \alpha) + \text{diag}(X) \nabla_X \alpha$ , so  $\sigma_{\min}(\partial Y / \partial X) \geq 1 - \|X\|_{\infty} \|\nabla_X \alpha\|_2$  by Weyl’s inequality [23]; under gate saturation ( $\nabla_X \alpha \rightarrow 0$ ),  $\sigma_{\min} \geq 1$  unconditionally.*

Figure 7. Per-block gate activation  $\alpha$  across CBA-MobileViT-S-lite (input source: local: sample.jpg)

 Fig. 2. Per-block gate activation  $\alpha = \sigma(z)$  from the trained CBA-MobileViT-S-lite checkpoint on an ImageNet-100 validation image. CBA-1 (at  $128 \times 128$  spatial extent) shows the most pronounced object-localised gate structure of the four blocks, consistent with the early receptive-field-expansion contribution (Section IV-D); deeper-stage blocks have progressively smaller and more uniform gate maps. Gate values are channel-averaged for visualisation.

 Fig. 3. Gate function  $\sigma(z)$  (dashed) and effective multiplier  $1 + \sigma(z)$  (solid blue) over  $z \in [-5, 5]$ , with the (1, 2) amplification band shaded. The  $D_2$  initialisation point  $z_{\text{init}} = -3$  gives  $1 + \sigma(-3) \approx 1.05$ , marked.

The gradient through a CBA block therefore cannot collapse, even at gate saturation — the multiplicative-gate analogue of He’16’s gradient-highway property [6].

**Proposition 3** (Bounded amplification). *With  $\alpha = \sigma(z) \in (0, 1)^d$ , the effective per-coordinate multiplier  $1 + \alpha_i \in (1, 2)$ , so  $|X_i| < |Y_i| < 2|X_i|$ .*

Signal is amplified through the block, never suppressed. The gate function is illustrated in Figure 3.

The lineage in the literature converges on the same pattern — gate the residual term, never the identity [19], [24], [25]. CBA’s per-pixel-per-channel choice generalises these (which gate per-channel or rank-1) to full  $C \times H \times W$  granularity.

**Initialisation.** The terminal  $D_2$  `project_bn.bias` is initialised to  $-3$ , giving  $\sigma(-3) \approx 0.047$  and effective multiplier  $1 + \alpha \approx 1.05$  at epoch 0 — block contributes  $\sim 5\%$  of identity at start and opens gradually (ReZero [25] warm-start principle). **DropPath** is applied per-sample to the gated residual only; the identity arm is never dropped. Linear schedule across the four CBA blocks: 0 at CBA-1 to

 TABLE I  
CBA BLOCK SPECIFICATIONS.

| Block | Input             | $C_{\text{mid}}$ | $C'$ | Latent           | $L$ | $t$ |
|-------|-------------------|------------------|------|------------------|-----|-----|
| CBA-1 | $32 \times 128^2$ | 48               | 72   | $16^2 \times 72$ | 2   | 4   |
| CBA-2 | $96 \times 32^2$  | 112              | 128  | $8^2 \times 128$ | 2   | 2   |
| CBA-3 | $128 \times 16^2$ | 144              | 160  | $8^2 \times 160$ | 3   | 2   |
| CBA-4 | $160 \times 8^2$  | 144              | 128  | $8^2 \times 128$ | 2   | 2   |

drop\_path\_rate at CBA-4.

#### H. CBA block: formal description

The block is summarised as Algorithm 1. Input is  $X \in \mathbb{R}^{B \times C \times H \times W}$ ; output is  $Y = X + \text{DropPath}(\alpha \odot X)$  with  $\alpha = \sigma(z)$  and  $z$  the output of the residual branch (encoder  $\rightarrow$  tokenize  $\rightarrow$  transformer  $\rightarrow$  de-tokenize  $\rightarrow$  decoder). Figure 4 traces the dataflow through one block; Figure 5 contrasts the resulting block against the MobileViT block it replaces.

### V. ARCHITECTURE: CBA-MOBILEVIT-S

#### A. Backbone (unchanged from MobileViT-S)

The full network is shown in Figure 6. The backbone is byte-for-byte identical to MobileViT-S [8]: stem (Conv  $3 \rightarrow 16$ ,  $3 \times 3$ , stride 2, BN, SiLU); Stage 1 ( $1 \times \text{MV2 } 16 \rightarrow 32$ ,  $t=1$ ,  $s=1$ ); Stage 2 ( $3 \times \text{MV2 } 32 \rightarrow 64$ ,  $t=4$ , first  $s=2$ ); Stage 3 ( $1 \times \text{MV2 } 64 \rightarrow 96$ ,  $t=4$ ,  $s=2$ ); Stage 4 ( $1 \times \text{MV2 } 96 \rightarrow 128$ ,  $t=4$ ,  $s=2$ ); Stage 5 ( $1 \times \text{MV2 } 128 \rightarrow 160$ ,  $t=4$ ,  $s=2$ ); head ( $1 \times 1$  Conv  $160 \rightarrow 640$ , BN, SiLU, AdaptiveAvgPool, FC  $640 \rightarrow 100$ ). Feature map sizes at  $256 \times 256$  input:  $128 \rightarrow 128 \rightarrow 64 \rightarrow 32 \rightarrow 16 \rightarrow 8$ .

#### B. CBA block instances

The four CBA blocks are sized for the  $\sim 2.8$  M-parameter budget along three axes: reduced bottleneck channels  $C'/C_{\text{mid}}$ , shallower deep stacks (CBA-3:  $L=3$ , CBA-4:  $L=2$ ), and hybrid expansion ratio ( $t=4$  at CBA-1,  $t=2$  at CBA-2/3/4). Per-block specifications are in Table I.

**Algorithm 1** CBA block forward pass (single block instance).

**Require:** Input feature map  $X \in \mathbb{R}^{B \times C \times H \times W}$ ; encoder strides  $(s_1, s_2)$ ; channel widths  $C, C_{\text{mid}}, C'$ ; expansion ratio  $t$ ; transformer depth  $L$ , head count  $h$ , FFN factor  $f=2$ ; DropPath rate  $p \in [0, 1)$ .

**Ensure:** Output feature map  $Y \in \mathbb{R}^{B \times C \times H \times W}$  with  $Y \approx X + (\sigma(z) \odot X)$ .

- 1:
- 2: **Encoder, stage 1** (MV2-style, stride  $s_1$ ):
- 3:  $z \leftarrow \text{PW}_{C \rightarrow tC}(X)$ ;  $z \leftarrow \text{SiLU}(\text{BN}(z))$   $\triangleright$  expand
- 4:  $z \leftarrow \text{DW}_{3 \times 3, s_1}(z)$ ;  $z \leftarrow \text{SiLU}(\text{BN}(z))$   $\triangleright$  spatial filter at expanded width
- 5:  $z \leftarrow \text{PW}_{tC \rightarrow C_{\text{mid}}}(z)$ ;  $z \leftarrow \text{BN}(z)$   $\triangleright$  linear project (Sandler §3)
- 6: **Encoder, stage 2** (MV2-style, stride  $s_2$ ):
- 7:  $z \leftarrow \text{PW} \circ \text{DW}_{3 \times 3, s_2} \circ \text{PW}_{\rightarrow C'}(z)$   $\triangleright$
- 8:  $z \in \mathbb{R}^{B \times C' \times h' \times w'}$ ,  $h' = H/(s_1 s_2)$
- 9: **Tokenize:**  $z \leftarrow \text{reshape}(z, B \times (h' w') \times C')$   $\triangleright$  spatial positions  $\rightarrow$  tokens
- 10: **Transformer ( $L$  layers, pre-norm):**
- 11: **for**  $\ell = 1$  **to**  $L$  **do**
- 12:  $z \leftarrow z + \text{MHSA}_h(\text{LN}(z))$   $\triangleright$  global mixing across  $h' w'$  tokens
- 13:  $z \leftarrow z + \text{FFN}_{f=2}(\text{LN}(z))$   $\triangleright$  rank-collapse counterweight (Dong'21, Yu'24)
- 14: **end for**
- 15: **De-tokenize:**  $z \leftarrow \text{reshape}(z, B \times C' \times h' \times w')$
- 16:
- 17: **Decoder, stage 1** (MV2-style + bilinear  $\times s_2$ ):
- 18:  $z \leftarrow \text{PW}_{C' \rightarrow tC'}(z)$ ;  $z \leftarrow \text{SiLU}(\text{BN}(z))$
- 19:  $z \leftarrow \text{Bilinear}_{\times s_2}(z)$ ;  $z \leftarrow \text{DW} \circ \text{PW}_{\rightarrow C_{\text{mid}}}(z)$ ;  $z \leftarrow \text{BN}(z)$
- 20: **Decoder, stage 2** (MV2-style + bilinear  $\times s_1$ , terminal):
- 21:  $z \leftarrow \text{PW} \circ \text{Bilinear}_{\times s_1} \circ \text{DW} \circ \text{PW}_{\rightarrow C}(z)$
- 22:  $z \leftarrow \text{BN}_{\beta=-3}(z)$   $\triangleright$  init:  $\sigma(-3) \approx 0.05 \Rightarrow$  block  $\approx$  identity
- 23:
- 24: **Gate and residual:**
- 25:  $\alpha \leftarrow \sigma(z) \in (0, 1)^{B \times C \times H \times W}$   $\triangleright$  per-pixel-per-channel gate
- 26:  $r \leftarrow \alpha \odot X$   $\triangleright$  Hadamard amplification
- 27: **if** training **and**  $p > 0$  **then**
- 28:  $r \leftarrow \text{DropPath}_p(r)$   $\triangleright$  stochastic depth on gated branch only
- 29: **end if**
- 30:  $Y \leftarrow X + r$   $\triangleright +X$  identity arm preserves He'16 gradient highway
- 31: **return**  $Y$

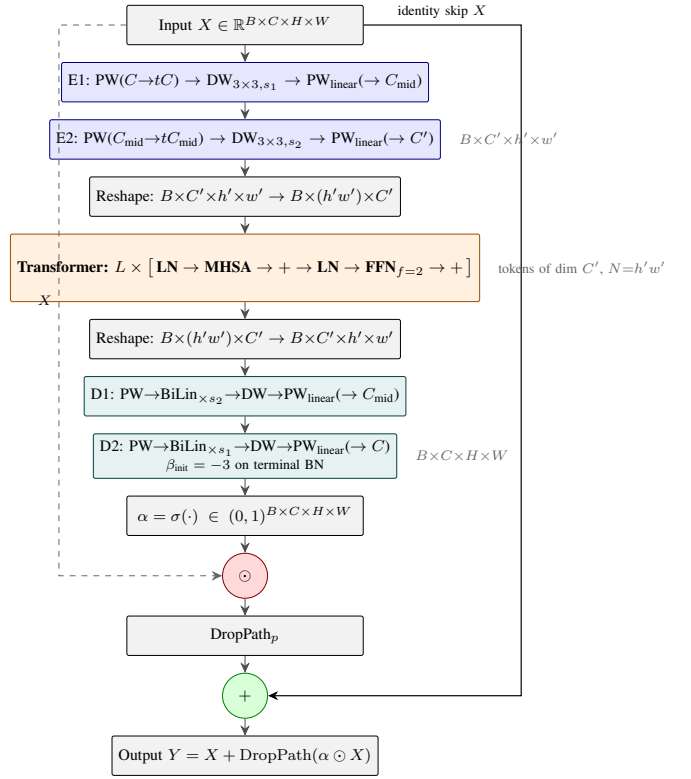


Fig. 4. CBA block internal architecture. The two-stage MV2-DSC encoder (E1, E2) compresses the input; the transformer reasons over the compressed latent; the symmetric two-stage decoder (D1, D2) maps the result back to the input shape. The terminal sigmoid produces the gate  $\alpha \in (0, 1)^{B \times C \times H \times W}$ , applied via Hadamard product inside an additive residual. The  $+X$  arm preserves the He'16 gradient highway (Proposition 2).

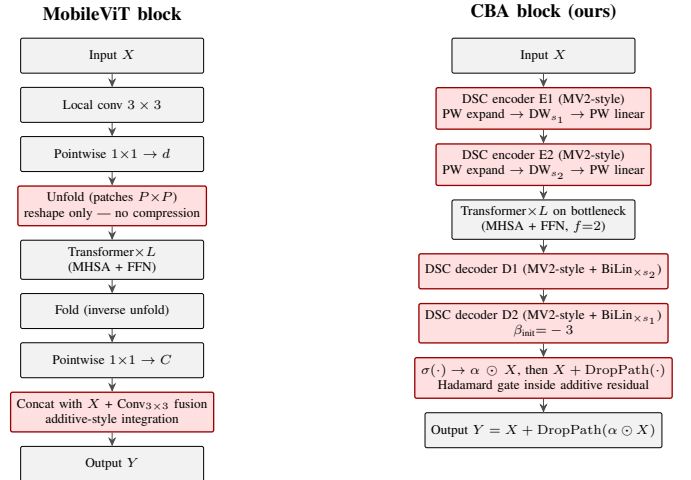


Fig. 5. Structural comparison of the MobileViT block (left) and the CBA block (right). Red-bordered boxes mark the differentiating components: MobileViT uses unfold/fold reshape (no spatial compression) followed by concat-fusion; CBA uses a two-stage non-linear DSC encoder/decoder around attention with Hadamard-gated additive residual integration.



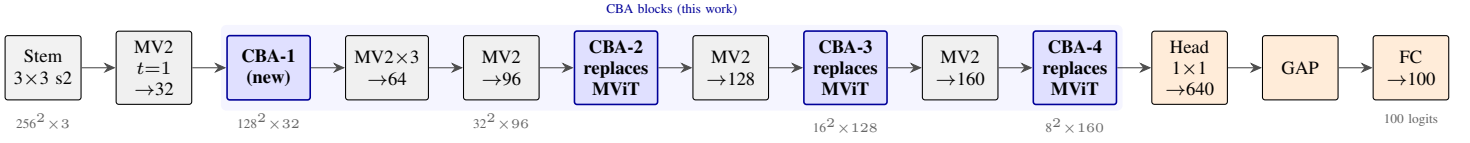


Fig. 6. CBA-MobileViT-S network skeleton. The convolutional backbone (stem + MV2 stages) is byte-for-byte identical to MobileViT-S [8]; the four CBA blocks (highlighted in blue) replace MobileViT-S’s three transformer blocks at stages 3/4/5 and add one new upstream block (CBA-1) after stage 1. CBA-1’s placement at  $128 \times 128$  spatial extent is the load-bearing instance of the early-receptive-field contribution (Section IV-D).

TABLE II  
AUTHORITATIVE PER-BLOCK MAC BREAKDOWN (MEASURED).

| Block                                       | MACs (M)      | Fraction      |
|---------------------------------------------|---------------|---------------|
| Stem (Conv 3 $\rightarrow$ 16, $s=2$ )      | 7.6           | 0.7%          |
| Stage 1 (MV2 16 $\rightarrow$ 32)           | 12.3          | 1.1%          |
| <b>CBA-1</b>                                | <b>297.5</b>  | <b>26.7%</b>  |
| Stage 2 (MV2 $\times 3$ , $\rightarrow$ 64) | 407.9         | 36.6%         |
| Stage 3 (MV2 $\rightarrow$ 96)              | 97.5          | 8.8%          |
| <b>CBA-2</b>                                | <b>93.6</b>   | <b>8.4%</b>   |
| Stage 4 (MV2 $\rightarrow$ 128)             | 52.3          | 4.7%          |
| <b>CBA-3</b>                                | <b>76.4</b>   | <b>6.9%</b>   |
| Stage 5 (MV2 $\rightarrow$ 160)             | 22.7          | 2.0%          |
| <b>CBA-4</b>                                | <b>39.3</b>   | <b>3.5%</b>   |
| Head $1 \times 1$ + AvgPool                 | 6.7           | 0.6%          |
| FC 640 $\rightarrow$ 100                    | 0.1           | 0.0%          |
| <b>Total</b>                                | <b>1113.7</b> | <b>100.0%</b> |

### C. Parameter and compute breakdown

The trained model has 2,785,956 parameters and 1.114 G MACs (fvcore) at  $256 \times 256$ ; Table II gives the per-block breakdown. Stage 2 (unchanged from MobileViT-S) contributes the largest single fraction at 36.6% but is shared with the baseline. The four CBA blocks contribute 45.5% together, dominated by **CBA-1** at 26.7% (about  $6\times$  the other three combined): CBA-1 sits at the highest spatial extent and runs a bilinear  $\times 4$  upsample at its terminal  $D_2$  stage, so the cost concentrates where the receptive-field-expansion contribution does.

## VI. EXPERIMENTAL SETUP

The recipe is the DeiT / MobileViTv2-family standard at 50-epoch length on ImageNet-100. Per-knob comparison across the three runs is in Table III.

**Optimizer.** AdamW [26], decoupled weight decay 0.05 ( $\text{ndim} \geq 2$  only), 10-epoch linear warmup then cosine to 0, gradient-clip at norm 1.0, label smoothing 0.1.

**Data and augmentation.** ImageNet-100 ( $\sim 126.7$  K train / 5 K val) at  $256 \times 256$ . RandomResizedCrop, horizontal flip, RandAugment ( $N=2, M=9$ ) [27], per-batch Mixup ( $\alpha=0.2$ ) [28] or CutMix ( $\alpha=1.0$ ) [29] with  $p=0.5$ .

**Regularisation.** Stochastic depth [30] on the gated branch only, linear  $0 \rightarrow 0.05$  across CBA-1... CBA-4. Model EMA decay 0.995; checkpoint is max(live, EMA).

**Hardware.** CBA-lite and MobileViT-XS: NVIDIA RTX PRO 6000 Blackwell, bf16 autocast,  $\sim 3\text{--}4.5$  h per run. MobileViTv2-0.75: Kaggle T4, fp16 + GradScaler,  $\sim 10$  h.

## VII. RESULTS

**CBA-MobileViT-S-lite reaches 81.52% top-1 (EMA) on ImageNet-100 at 50 epochs** —  $+4.62$  pp over the matched-protocol MobileViT-XS control at  $+0.18$  G MACs, and  $+3.48$  pp over the asymmetric MobileViTv2-0.75 reference at  $+0.06$  G MACs. All runs trained from scratch at  $256 \times 256$ . CBA-lite and MobileViT-XS are byte-for-byte parity-locked (Blackwell, B=512, peak LR  $1.5 \times 10^{-3}$ , decoupled-WD AdamW, 10-epoch warmup + cosine, RandAug + Mixup/CutMix, drop\_path 0.05, EMA decay 0.995, seed 42); the v2-0.75 run used a different schedule on a Kaggle T4 (disclosed below).

### A. Headline results

The MAC overheads are  $+19\%$  over MobileViT-XS and  $+6\%$  over MobileViTv2-0.75 — both small relative to the accuracy gaps. The accuracy-vs-MACs Pareto plot is Figure 7.

**On the v2-0.75 control.** The MobileViTv2-0.75 run used batch size 128 with peak LR  $4 \times 10^{-4}$  on a Kaggle T4 —  $\sim 4\times$  more optimizer steps per epoch and a marginally higher LR-to-batch ratio than CBA-lite. Both effects favour v2-0.75 under a fixed 50-epoch schedule, so the  $+3.48$  pp CBA-lite advantage is understated, not overstated.

**On deployment scope.** The accuracy / parameter / theoretical-FLOPs comparison is the strongest claim the available compute supports. Wall-clock latency on mobile CPU, NPU, or MCU is not measured — edge-deployment testing requires runtime export, per-device tuning, and hardware access beyond the project budget. The headline is a *theoretical* efficiency result; deployment evaluation is deferred (Section VIII-D).

### B. Training trajectory

Figure 8 plots per-epoch curves for all three runs. The CBA-lite variant trains cleanly: no warmup-spike events, monotonic loss decrease through epoch 10, and a val/EMA gap that stays within the expected 0–4 pp range. The EMA model tracks live within  $\pm 0.2$  pp by mid-training.

## VIII. DISCUSSION

### A. What the result establishes

CBA’s claim is that **deep bottleneck attention belongs at high spatial extent**. The 81.52% top-1 ( $+4.62$  pp over the matched MobileViT-XS) is the feasibility result; the architectural payoff is full receptive field at a stage where pure

TABLE III

TRAINING-SETUP COMPARISON ACROSS THE THREE LITE-BUDGET RUNS. GREEN = MATCHED ACROSS ALL THREE; YELLOW = MATCHED BETWEEN CBA-LITE AND MOBILEViT-XS ONLY (THE DISCLOSED MOBILEViTv2-0.75 ASYMMETRY).

Training-setup comparison — matched protocol vs. disclosed asymmetries

| Setting                  | CBA-MobileViT-S-lite (ours) | MobileViT-XS           | MobileViTv2-0.75     | Status                            |
|--------------------------|-----------------------------|------------------------|----------------------|-----------------------------------|
| Hardware (GPU)           | RTX PRO 6000 Blackwell      | RTX PRO 6000 Blackwell | Kaggle (Tesla-class) | Matched (CBA↔XS); v2-0.75 differs |
| Resolution               | 256×256                     | 256×256                | 256×256              | Matched (all three)               |
| Dataset                  | ImageNet-100                | ImageNet-100           | ImageNet-100         | Matched (all three)               |
| Epochs                   | 50                          | 50                     | 50                   | Matched (all three)               |
| Batch size               | 512                         | 512                    | 128                  | Matched (CBA↔XS); v2-0.75 differs |
| Steps / epoch            | 247                         | 247                    | ~990                 | Matched (CBA↔XS); v2-0.75 differs |
| Peak LR                  | 1.5e-3                      | 1.5e-3                 | 4.0e-4               | Matched (CBA↔XS); v2-0.75 differs |
| LR / batch ratio         | 2.93e-6                     | 2.93e-6                | 3.13e-6              | Matched (CBA↔XS); v2-0.75 differs |
| Warmup epochs            | 10                          | 10                     | 5                    | Matched (CBA↔XS); v2-0.75 differs |
| LR schedule              | cosine to 0                 | cosine to 0            | cosine to 0          | Matched (all three)               |
| Optimizer                | AdamW                       | AdamW                  | AdamW                | Matched (all three)               |
| Weight decay (decoupled) | 0.05                        | 0.05                   | 0.05                 | Matched (all three)               |
| Gradient clip (norm)     | 1.0                         | 1.0                    | 1.0                  | Matched (all three)               |
| Label smoothing          | 0.1                         | 0.1                    | 0.1                  | Matched (all three)               |
| RandAugment (N, M)       | (2, 9)                      | (2, 9)                 | (2, 9)               | Matched (all three)               |
| Mixup $\alpha$           | 0.2                         | 0.2                    | 0.2                  | Matched (all three)               |
| CutMix $\alpha$          | 1.0                         | 1.0                    | 1.0                  | Matched (all three)               |
| Mixup/CutMix prob        | 0.5 (50/50 split)           | 0.5 (50/50 split)      | 0.5 (50/50 split)    | Matched (all three)               |
| Stochastic depth         | 0.05                        | 0.05                   | 0.05                 | Matched (all three)               |
| EMA decay                | 0.995                       | 0.995                  | 0.999                | Matched (CBA↔XS); v2-0.75 differs |
| Mixed precision          | bf16                        | bf16                   | bf16                 | Matched (all three)               |
| Random seed              | 42                          | 42                     | 42                   | Matched (all three)               |
| Dataloader workers       | 8                           | 8                      | 8                    | Matched (all three)               |

Matched (all three) Matched (CBA↔XS); v2-0.75 differs Differs (all three)

TABLE IV

LITE-BUDGET EVALUATION ON IMAGENET-100, 50 EPOCHS, 256×256. *Best top-1* is max(LIVE, EMA).

| Model                              | Params        | MACs (fvcore)  | Val top-1 | Val top-5 | EMA top-1     | Best top-1    |
|------------------------------------|---------------|----------------|-----------|-----------|---------------|---------------|
| <b>CBA-MobileViT-S-lite (ours)</b> | <b>2.79 M</b> | <b>1.114 G</b> | 81.32%    | 95.76%    | <b>81.52%</b> | <b>81.52%</b> |
| MobileViT-XS [8]                   | 1.97 M        | 0.935 G        | 76.78%    | 94.52%    | 76.78%        | 76.90%        |
| MobileViTv2-0.75 [9]               | 2.52 M        | 1.051 G        | 77.90%    | 94.40%    | 77.86%        | 78.04%        |

CNN delivers  $\sim 11$ -pixel local features. The Hadamard-gated additive residual is a supporting choice, not the contribution.

B. Where the architecture should and shouldn't help

CBA should help where evidence from distant spatial positions matters early — fine-grained recognition, scene-conditioned classification, occlusion-heavy benchmarks, and corruption robustness — and Figure 2 shows compute concentrating on object regions consistent with this. It should *not* help where local features are already sufficient (coarse texture, centred objects, micro-images), and the budget is fixed: CBA-1 alone consumes 26.7% of total MACs, so dynamic-token-pruning [16], [18] retains a flexibility advantage there.

C. Limitations

- **ImageNet-100 only.** Generalisation to IN-1K, longer schedules, and downstream tasks (segmentation, detection) is future work.

- **Single-seed numbers.** Single-seed variance at this scale is typically 0.3–0.7 pp; the +4.62 pp / +3.48 pp gaps sit comfortably outside that band, but a 3-seed bound is the publishable evidence.
- **Single-budget evaluation.** A larger-budget variant and its matched MobileViTv2-1.0 control are deferred to a compute-funded extension.
- **No 2024-class hybrid baseline.** EdgeNeXt [31], FastViT [11], EfficientFormerV2 [10], RepViT [12] are not in the headline.
- **No edge-device deployment.** The efficiency claim is *theoretical*; wall-clock latency is out of scope.
- **No dense-prediction transfer.** Segmentation (e.g., ADE20K [32]) and detection are future work.

D. Future work

The highest-priority follow-ups: (1) multi-seed re-runs of the headline ( $\sim 9$  GPU-hours, 3 seeds); (2) the *CBA-1 ablation* (drop the early block, keep CBA-2/3/4) — the cleanest test



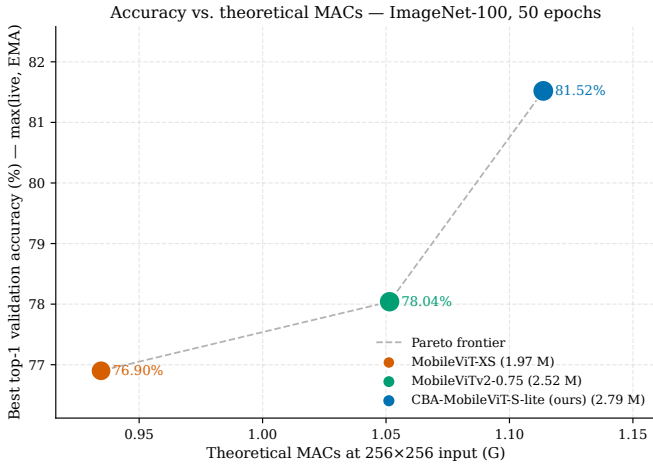


Fig. 7. Accuracy vs. theoretical MACs. ImageNet-100, 50 epochs.  $y$ -axis: best top-1 = max(live, EMA). Point area proportional to parameter count.

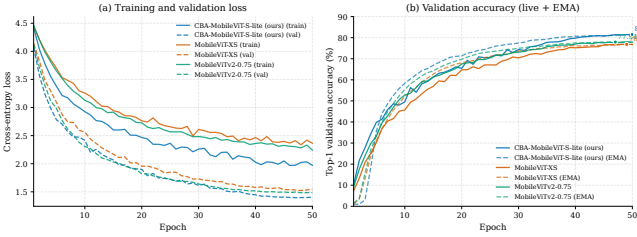


Fig. 8. Training trajectory across 50 epochs. Left: train loss (solid) and validation loss (dashed) for all three lite-budget runs. Right: top-1 validation accuracy (live) and EMA (dashed); best EMA points annotated.

of the early-receptive-field claim of Section IV-D; (3) the inverse *CBA-only-1* ablation (keep only the early block); (4) a latent-geometry probe on CBA-1’s encoder output (within- vs. between-class cosine similarity and linear-probe accuracy), converting the open question of Section IV-B into measured data; (5) noFFN and additive ablations testing Sections IV-F and IV-G. Beyond ablations: edge-device deployment (TFLite / Core ML / ONNX-Runtime), a larger-budget variant against MobileViTV2-1.0, comparison against a 2024-class hybrid baseline (e.g., EdgeNeXt-XS [31]), knowledge distillation, and downstream transfer to segmentation (ADE20K [32]) and detection (COCO [33]).

## IX. CONCLUSION

Hybrid mobile vision transformers defer attention to the late stages of the network, leaving early stages pure-convolutional with locally-bounded receptive field. **Deep bottleneck attention** is the natural alternative; the structural concerns that left it unoccupied — latent geometry and compute cost — are addressed by treating the first as an open empirical question grounded in end-to-end self-consistency, and the second by convolutional operators in the MobileNetV2-style depthwise-separable variant. The architectural payoff is

full receptive field at the  $128 \times 128$  output of MobileViT-S’s stage 1, where standard hybrids still operate on  $\sim 11$ -pixel local patches. CBA-MobileViT-S-lite (2.79 M params, 1.114 G MACs) trained on ImageNet-100 reaches 81.52% **top-1 (EMA)** — +4.62 pp over the matched MobileViT-XS control and +3.48 pp over MobileViTV2-0.75. The 2024-frontier positioning, multi-seed bound, and edge-deployment evaluation are extensions catalogued in Section VIII-D.

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